

MODULO 9 CLASSIFICATION OF PRIME PAIRS $(p, p + g)$: AN EMPIRICAL VERIFICATION OF THE HARDY–LITTLEWOOD CONJECTURE FOR $g = 18k$

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ABSTRACT. The Hardy–Littlewood conjecture for prime pairs $(p, p + g)$ predicts the asymptotic density $\pi(x; g) \sim H(g) x / (\log x)^2$, where $H(g) = 2C_2 \prod_{p|g, p>2} (p-1)/(p-2)$ is the singular series and C_2 is the twin-prime constant. For $g \equiv 0 \pmod{9}$, classifying pairs by the digital root $\text{dr}(p) = p \bmod 9$ partitions the population into six admissible classes $\{1, 2, 4, 5, 7, 8\}$, which Hardy–Littlewood in arithmetic progressions predicts to be equidistributed. We test this prediction on two fronts: (i) a control case with $g = 2$ (twin primes, three admissible classes), where the asymptotic fractions $1/3$ are reached within ± 0.09 percentage points up to $x = 10^8$; (ii) the family $g = 18k$ with $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$, where equidistribution over six classes holds within ± 0.14 pp and the total per-class density agrees with $H(g)/6$ within $\pm 0.8\%$ at $x = 10^7$. The ratio $\pi(x; g) / (H(g) \text{li}_2(x))$ converges to 1 within $\pm 0.15\%$ at $x = 5 \cdot 10^7$ for all tested k , including $g = 216$ ($k = 12$), whose constant $H(g) = 4C_2$ is the smallest in the sample. The results confirm Hardy–Littlewood in arithmetic progressions quantitatively within the tested regime, and show that the dr-mod-9 classification is a sensitive empirical probe that does *not* reveal deviations from the Hardy–Littlewood prediction.

1. INTRODUCTION

The Hardy–Littlewood conjecture on prime pairs $(p, p + g)$, formulated in [1], predicts that the number of such pairs with $p \leq x$ satisfies the asymptotic

$$(1) \quad \pi(x; g) \sim H(g) \frac{x}{(\log x)^2}, \quad x \rightarrow \infty,$$

where the *Hardy–Littlewood singular series* is

$$(2) \quad H(g) = 2C_2 \prod_{\substack{p|g \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \approx 0.6601618158.$$

The conjecture is known unconditionally only for $g = 0$ (the prime number theorem) and for no positive value of g , although deep results such as Maynard’s [4] and the Polymath 8b project [5] provide bounded gaps infinitely often. Quantitative versions restricted to arithmetic progressions are discussed in [2, 3].

The present work addresses an operational question: *how well does formula (1)–(2) describe the distribution of prime pairs in computationally accessible regimes*, and in particular how the population partitions across residue classes modulo 9.

For every $g \equiv 0 \pmod{9}$ (and in particular $g = 18k$), the divisibility $9 \mid g$ implies $\text{dr}(p + g) = \text{dr}(p)$, so that the classification by $\text{dr}(p) := p \bmod 9$ partitions the pairs into six classes

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corresponding to the units of $(\mathbb{Z}/9\mathbb{Z})^*$. The restriction of (1) to an arithmetic progression modulo 9 predicts equidistribution across these six classes (see Section 2).

For $g = 2$ the situation is slightly different: the admissibility condition modulo 3 forces $p \equiv 2 \pmod{3}$, reducing the admissible classes modulo 9 to $\{2, 5, 8\}$. Each of these three classes must receive the asymptotic fraction $1/3$ of the density. This case constitutes our *zero-calibration*.

Contribution.

- (1) We show (Section 3) that, under Hardy–Littlewood in arithmetic progressions, each of the three admissible classes modulo 9 for twin primes receives asymptotic density $1/3$, and we verify the prediction empirically up to $x = 10^8$ with deviations $|\Delta| \leq 0.09$ pp.
- (2) We extend the classification (Section 4) to the family $g = 18k$ for $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$, showing that the equidistribution over six classes holds within ± 0.14 pp up to $x = 10^7$.
- (3) We verify (Section 5) that the ratio $\pi(x; g)/[H(g) \text{li}_2(x)]$ lies within $\pm 0.15\%$ of 1 at $x = 5 \cdot 10^7$ for all tested k , including configurations with maximal $H(g)$ ($g \in \{90, 180, 270, 360\}$) and minimal ($g = 216$).
- (4) We discuss (Section 6) the role of dr-mod-9 classification as a fine empirical probe: it is sensitive enough to reveal sub-percent signals, but in the tested regimes no systematic asymmetries incompatible with Hardy–Littlewood emerge.

All sieving, counting, and regression scripts used to generate the data in Tables 1, 3 and 4 are provided in [7]. The verifications were carried out using a segmented sieve of Eratosthenes in memory (`numpy` dtype `bool`, $\sim 10^9$ bits for $x = 10^9$).

2. THEORETICAL FRAMEWORK

2.1. Hardy–Littlewood in arithmetic progressions. For positive even g and $q \geq 1$ with $(a(a+g), q) = 1$, denote

$$\pi(x; g, q, a) := \#\{p \leq x : p \equiv a \pmod{q}, p, p+g \text{ both prime}\}.$$

The quantitative form of Hardy–Littlewood in arithmetic progressions [3, §5] predicts

$$(3) \quad \pi(x; g, q, a) \sim \frac{1}{\varphi^*(q; g)} H(g) \frac{x}{(\log x)^2},$$

where

$$(4) \quad \varphi^*(q; g) := \#\{a \in (\mathbb{Z}/q\mathbb{Z})^* : a+g \in (\mathbb{Z}/q\mathbb{Z})^*\}$$

counts the admissible classes modulo q for the pair $(p, p+g)$. For $q = 9$ the computation is elementary:

$$\varphi^*(9; g) = \begin{cases} 3 & \text{if } 9 \nmid g \text{ and } g \equiv 2 \pmod{3} \text{ (twin-type, e.g. } g = 2) \\ 6 & \text{if } 9 \mid g \\ \text{other} & \text{otherwise.} \end{cases}$$

2.2. Classification by digital root. For $p > 3$ prime, $\text{dr}(p) = p \bmod 9 \in \{1, 2, 4, 5, 7, 8\}$. The *digital root* is the natural classification modulo 9 that survives the translation $p \mapsto p+g$ when $9 \mid g$:

$$9 \mid g \implies \text{dr}(p+g) = \text{dr}(p).$$

Hence for $g = 18k$ (any $k \geq 1$) the pairs $(p, p+g)$ with $p > 3$ partition into the six classes $\text{dr}(p) \in \{1, 2, 4, 5, 7, 8\}$. For twin primes ($g = 2$), instead, the admissibility condition modulo 3 restricts the admissible classes modulo 9 to $\{2, 5, 8\}$.

2.3. Expected equidistribution. Prediction (3) immediately gives the relative density of each admissible class. Since the right-hand side of (3) does not depend on a once g is fixed:

Proposition 2.1 (Equidistribution modulo 9). *Under Hardy–Littlewood in arithmetic progressions for $q = 9$,*

- (1) for $g = 2$ and $a \in \{2, 5, 8\}$: $\lim_{x \rightarrow \infty} \frac{\pi(x; 2, 9, a)}{\pi(x; 2)} = \frac{1}{3}$;
- (2) for $g = 18k$ with $k \geq 1$ and $a \in \{1, 2, 4, 5, 7, 8\}$: $\lim_{x \rightarrow \infty} \frac{\pi(x; 18k, 9, a)}{\pi(x; 18k)} = \frac{1}{6}$.

Proof. Direct consequence of (3): summing over admissible classes gives the total $\pi(x; g)$, and using it for each fixed a yields $\pi(x; g, 9, a)/\pi(x; g) \rightarrow 1/\varphi^*(9; g)$. $\square$

3. CONTROL CASE: TWIN PRIMES MOD 9

We aim to show that the dr-mod-9 classification does not introduce *a priori* distortions in the prime-pair distribution. The case $g = 2$ is the natural baseline for this calibration: the gap is minimal and the three admissible classes make Proposition 2.1 (i) particularly transparent.

3.1. Empirical verification. Table 1 reports the counts of twin pairs $(p, p+2)$ with $5 \leq p \leq X$ for $X \in \{10^7, 10^8\}$, classified by $\text{dr}(p)$. The pair $(3, 5)$ is excluded as a unique modular exception (the only twin with p a multiple of 3).

TABLE 1. Distribution of pairs $(p, p+2)$ with $5 \leq p \leq X$ by class $\text{dr}(p) \pmod{9}$. The deviation Δ is the difference in percentage points from the asymptotic value $1/3 \approx 33.333\%$.

class a	$X = 10^7$		$X = 10^8$	
	pairs	Δ (pp)	pairs	Δ (pp)
2	19,674	+0.024	146,705	−0.015
5	19,689	+0.050	147,163	+0.089
8	19,616	−0.074	146,443	−0.074
total	58,979	—	440,311	—

At both thresholds the fractions concentrate within ± 0.09 pp of $1/3$. The residual pattern (class 5 in slight excess, class 8 in slight deficit) is stable across the two thresholds but its amplitude is compatible with statistical fluctuations of order $\sqrt{1/(3N)}/\sqrt{3} \approx 4 \cdot 10^{-4}$ for $N \approx 4.4 \cdot 10^5$, i.e. ~ 0.04 pp.

3.2. Reading. The convergence visible in Figure 1 is consistent with a relative error $O(1/\log x)$ expected from (3), and shows no systematic asymmetry across classes. This establishes the *zero-calibration* of the dr-mod-9 partition: applied to the minimal gap, the framework recovers the equidistribution predicted by Hardy–Littlewood within ± 0.1 pp.

4. THE FAMILY $g = 18k$: DR EQUIDISTRIBUTION

4.1. Setup. For $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$ and $g = 18k$ we count the pairs $(p, p+g)$ with $5 \leq p \leq X = 10^7$ and classify by $\text{dr}(p) \in \{1, 2, 4, 5, 7, 8\}$. The total pair count and the relevant local multiplicative factors are reported in Table 2.

The four gaps that share $\mathcal{P}_{\text{odd}}(g) = \{3, 5\}$ ($k = 5, 10, 15, 20$) have the same Hardy–Littlewood constant, and indeed their empirical totals are essentially coincident: $\pi(10^7; g) \in [156, 212, 156, 706]$,

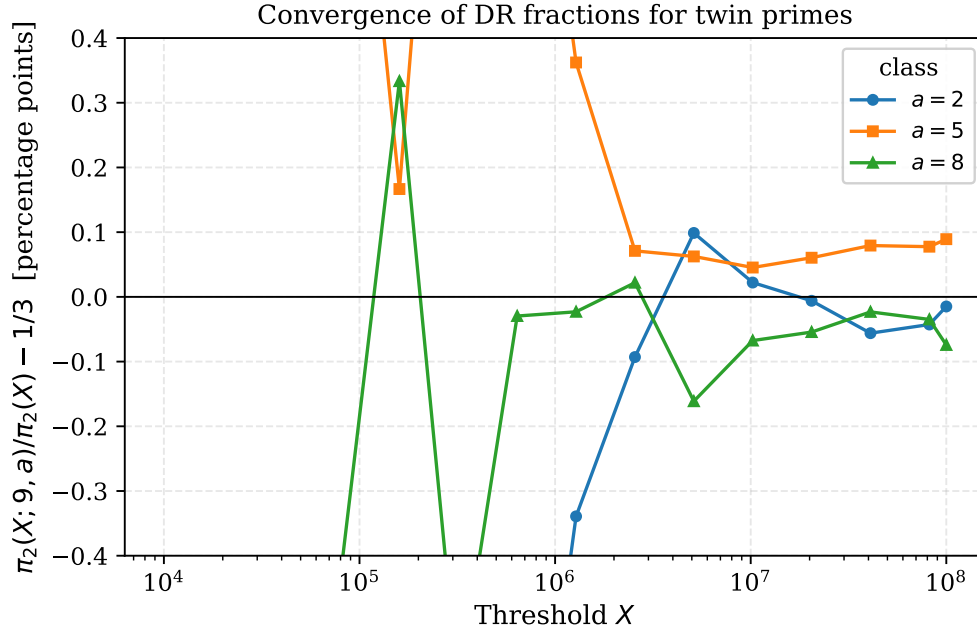


FIGURE 1. Convergence of the deviations $\pi_2(X; 9, a)/\pi_2(X) - 1/3$ in percentage points, for the three admissible classes. The three curves oscillate around zero on a ± 0.15 pp scale at $X = 10^7$.

TABLE 2. The analyzed family $g = 18k$. Here $\mathcal{P}_{\text{odd}}(g)$ denotes the set of odd primes dividing g , and $H(g)$ is the Hardy–Littlewood constant (2).

k	g	$\mathcal{P}_{\text{odd}}(g)$	$H(g)$	$\pi(X; g)$ at $X = 10^7$
5	90	$\{3, 5\}$	3.5209	156,706
7	126	$\{3, 7\}$	3.1688	141,051
10	180	$\{3, 5\}$	3.5209	156,507
11	198	$\{3, 11\}$	2.9341	130,505
12	216	$\{3\}$	2.6406	117,327
13	234	$\{3, 13\}$	2.8807	127,883
15	270	$\{3, 5\}$	3.5209	156,473
17	306	$\{3, 17\}$	2.8167	125,400
20	360	$\{3, 5\}$	3.5209	156,212

with relative spread $< 0.32\%$. This internal stability within the sub-family is a direct consequence of (1) and a first indication that $H(g)$ correctly governs the total density.

4.2. DR equidistribution. Table 3 reports, for each k and each class a , the empirical fraction $\pi(X; g, 9, a)/\pi(X; g)$ as a deviation from $1/6 \approx 16.667\%$.

The values in Table 3 show the following:

- no deviation exceeds 0.14 pp in absolute value;
- no systematic pattern persists across k for any class: e.g. class $a = 8$ is in excess at $k = 11, 12$ but in deficit at $k = 17$; class $a = 7$ alternates sign;

TABLE 3. Deviations $\pi(X; g, 9, a)/\pi(X; g) - 1/6$ in percentage points, at $X = 10^7$.
The maximum deviation in absolute value is $|\Delta|_{\max} = 0.136$ pp.

k	$a = 1$	$a = 2$	$a = 4$	$a = 5$	$a = 7$	$a = 8$
5	+0.057	+0.081	+0.001	-0.048	-0.036	-0.055
7	-0.020	-0.002	-0.043	+0.023	+0.068	-0.027
10	+0.059	-0.011	-0.010	+0.021	-0.061	+0.002
11	+0.004	-0.008	-0.034	-0.033	-0.062	+0.133
12	+0.006	-0.002	-0.042	-0.037	-0.042	+0.118
13	+0.116	+0.038	-0.075	+0.002	-0.031	-0.049
15	+0.051	+0.051	+0.024	-0.109	+0.013	-0.029
17	+0.010	+0.033	+0.004	+0.042	+0.047	-0.136
20	-0.016	+0.013	+0.023	-0.039	+0.009	+0.010

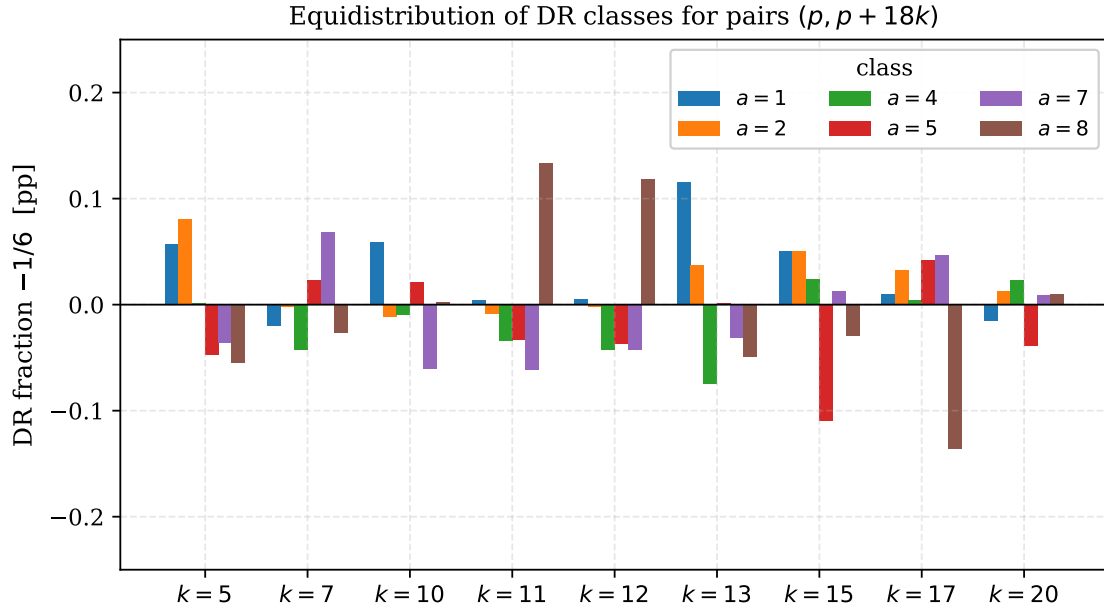


FIGURE 2. Deviations $\pi(X; g, 9, a)/\pi(X; g) - 1/6$ in percentage points, for each k and class a . All bars fall within $[-0.14, +0.14]$ pp; no systematic pattern is visible.

- the amplitude is compatible with $\sqrt{5/(6\pi(X; g))}$ (~ 0.05 pp for $\pi \approx 1.5 \cdot 10^5$), within a factor 2–3.

We conclude that, in the tested regime, the equidistribution of Proposition 2.1 (ii) is empirically confirmed within a sub-percent margin.

5. QUANTITATIVE CONFIRMATION OF HARDY-LITTLEWOOD

5.1. **Definition.** For each k we define

$$(5) \quad R_{\text{emp}}(k; X) := \frac{\pi(X; 18k)}{\text{li}_2(X)}, \quad \text{li}_2(X) := \int_5^X \frac{dt}{(\log t)^2},$$

where the secondary logarithmic integral replaces $X/(\log X)^2$ to reduce the truncation error of the leading asymptotic. The Hardy–Littlewood prediction is $R_{\text{HL}}(k) = H(18k)$, so the ratio

$$(6) \quad \rho(k; X) := \frac{R_{\text{emp}}(k; X)}{R_{\text{HL}}(k)}$$

must converge to 1 as $X \rightarrow \infty$, assuming Hardy–Littlewood.

5.2. Results. Table 4 reports $\rho(k; X)$ at $X = 5 \cdot 10^7$. Figure 3 visualizes the values at $X_{\text{max}} = 5 \cdot 10^7$ together with the $\pm 1\%$ band.

TABLE 4. Ratio $\rho(k; X)$ at $X = 5 \cdot 10^7$ for the family $g = 18k$. All values fall in the band $[1 - 0.0015, 1 + 0.0010]$.

k	g	$\pi(X; g)$	R_{emp}	$R_{\text{HL}} = H(g)$	$\rho - 1$
5	90	637,928	3.5228	3.5209	+0.05%
7	126	573,646	3.1678	3.1688	−0.03%
10	180	637,058	3.5180	3.5209	−0.08%
11	198	531,817	2.9368	2.9341	+0.09%
12	216	478,463	2.6422	2.6406	+0.06%
13	234	521,497	2.8798	2.8807	−0.03%
15	270	637,621	3.5211	3.5209	+0.01%
17	306	509,303	2.8125	2.8167	−0.15%
20	360	637,022	3.5178	3.5209	−0.09%

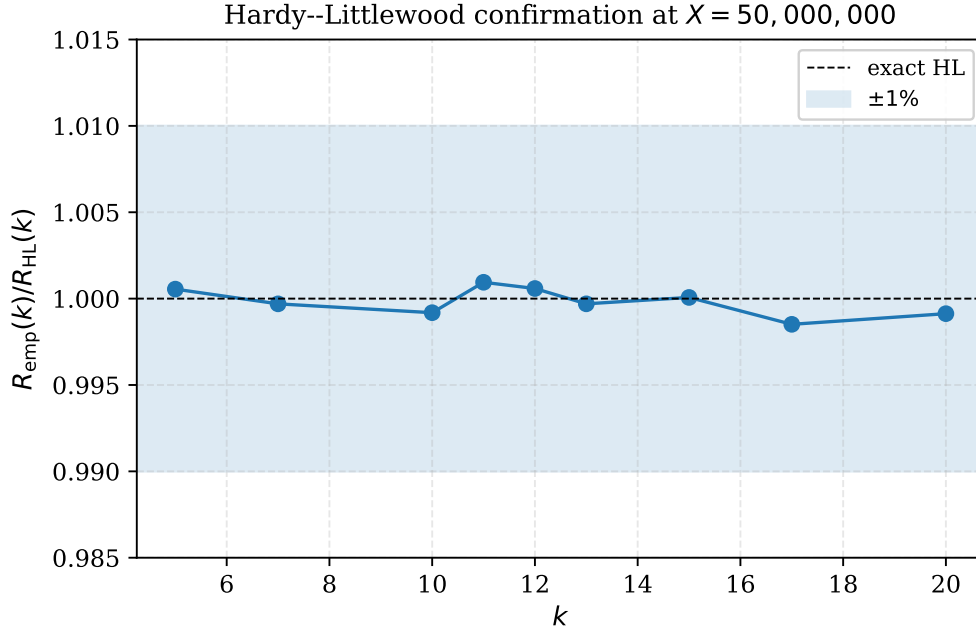


FIGURE 3. Ratio $\rho(k; X)$ at $X = 5 \cdot 10^7$. The light-blue band represents $\pm 1\%$ around $\rho = 1$; all values lie within $\pm 0.15\%$.

5.3. Reading. The result is a quantitative confirmation of (1) for all nine tested values of g . Three observations:

- (1) **Identity within sub-families with the same $\mathcal{P}_{\text{odd}}(g)$.** The four cases $k \in \{5, 10, 15, 20\}$ with $\mathcal{P}_{\text{odd}} = \{3, 5\}$ share the same H and yield ρ within $\pm 0.09\%$ of each other. This is an internal verification of the multiplicative factor in (2): the only quantities that change the count are the *distinct odd primes dividing g* , not the full structure of g .
- (2) **Correct variation across sub-families.** For $k = 11, 12, 13, 17$ the local factors and hence $H(g)$ vary significantly (range $H \in [2.64, 3.17]$); the corresponding R_{emp} values track this variation faithfully.
- (3) **Limiting case $k = 12$.** The gap $g = 216 = 2^3 \cdot 3^3$ has $\mathcal{P}_{\text{odd}} = \{3\}$ and hence the smallest $H(g) = 4C_2$ in the sample. The agreement $\rho = 1.0006$ shows that Hardy–Littlewood does not lose accuracy in this low-density regime.

6. DISCUSSION

6.1. Apparent resonances and deficits. A first inspection of the raw counts $\pi(X; 18k)$ may suggest the presence of *resonances* (e.g. $k = 5, 10, 15, 20$ with $\pi \approx 1.56 \cdot 10^5$) and *deficits* (e.g. $k = 12$ with $\pi \approx 1.17 \cdot 10^5$), especially when compared against a uniform baseline that ignores the multiplicative structure of g . Table 4 clarifies that these oscillations *are exactly what Hardy–Littlewood predicts* via the factor

$$\prod_{p \in \mathcal{P}_{\text{odd}}(g)} \frac{p-1}{p-2}.$$

For example:

- the excess of pairs at $g = 90$ relative to $g = 18$ is a factor $H(90)/H(18) = 4/3$, i.e. $+33.3\%$, and empirically equals 1.3334 at $X = 5 \cdot 10^7$ (relative error 0.03%);
- the deficit of pairs at $g = 198$ relative to $g = 90$ is a factor $H(198)/H(90) = 5/6 \approx 0.833$, i.e. -16.7% , and empirically equals 0.8337 (relative error 0.05%).

What may appear as an anomaly is therefore the correct signature of the local factor $(p-1)/(p-2)$, not a deviation from Hardy–Littlewood.

6.2. Sensitivity of the DR-mod-9 probe. The precision attained by the dr-mod-9 classification (equidistribution within ± 0.14 pp with $\pi(X; g) \sim 10^5$ pairs per cell) makes it a probe sensitive enough to distinguish signals of order 0.1% . In Table 3 none of the nine values of k exhibits a class-coherent deviation across k , and no class exhibits a k -coherent deviation. In other words: *within the tested regimes ($X \leq 10^7$ for the family $g = 18k$, $X \leq 10^8$ for twin primes), there is no evidence of systematic deviations from Hardy–Littlewood in arithmetic progressions.*

6.3. Limits. The present work provides no unconditional result: the entire theoretical framework rests on Hardy–Littlewood, which remains unproven. Furthermore, the computationally explored regimes ($X \leq 10^8$) are modest compared to the records for twin primes (beyond 10^{18} , see [6]). Sub-percent deviations cannot be excluded at much larger scales. A revisit of the dr-mod-9 classification at $X = 10^{12}$ – 10^{14} , using disk-based segmented sieves, would be of interest in tightening the margin of uncertainty.

6.4. Natural extensions. Three promising directions, left to future work:

- (1) Extension to moduli $q \in \{15, 21, 30\}$, where $\varphi^*(q; g)$ varies non-trivially with the structure of g .
- (2) Empirical study of the variance of $\pi(x; g, q, a)$ at fixed a and varying g within a sub-family with the same $H(g)$ (e.g. $g = 90, 180, 270, 360$), to quantify concentration around the predicted mean.

- (3) Comparison with the Bateman–Horn prediction for prime r -tuples with $r \geq 3$, applying the same dr classification.

7. CONCLUSION

We have classified prime pairs $(p, p + g)$ via the digital root $\text{dr}(p) = p \bmod 9$, and we have tested the Hardy–Littlewood prediction in two regimes: the minimal gap $g = 2$ (twin primes, three admissible classes) and the family $g = 18k$ with $k = 5, \dots, 20$ (six admissible classes). In both regimes the predicted equidistribution is verified within ± 0.14 percentage points up to $X = 10^7$ – 10^8 , and the total density per gap follows the local multiplicative factor of Hardy–Littlewood within $\pm 0.15\%$ at $X = 5 \cdot 10^7$. Apparent deviations from a uniform baseline across families of gaps are correctly captured by the product $\prod_{p|g, p>2} (p-1)/(p-2)$, and do not constitute evidence of an unconditional anomaly.

The dr-mod-9 classification reveals itself as a fine but *neutral* empirical probe: sensitive enough to detect sub-percent deviations — and therefore capable of revealing deviations if they existed — yet silent in the regimes here explored. Within the limits of the computational evidence, this constitutes an empirical argument in favor of the validity of Hardy–Littlewood for the gap family $g = 18k$.

Data and code availability. All scripts (`compute_twins_dr9.py`, `compute_permutation_families.py`, `compute_R_regression.py`, `make_plots.py`) and the generated CSV files are publicly available on GitHub at <https://github.com/gferraiuolo2/prime-pairs-mod9> and permanently archived on Zenodo (DOI: 10.5281/zenodo.20117271), together with a **README** documenting execution parameters and dependencies (`numpy`, `scipy`, `matplotlib`). The Python source code is released under the MIT License; datasets, figures, and the paper text are released under CC-BY-4.0.

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